

Synthesized Spatial Intelligence

Integrating the God's Eye View Engine with GIS & YOLO Edge AI



[SYS.01: GEV 3D VISUAL ENGINE [ONLINE]]

[STATUS: RENDER_OK]
[RES: 8K ULTRA_HD]
[FPS: 60 STABLE]
[FOV: WIDE_ANGLE]
[GEO_DATA: INTEGRATED]
[LATENCY: <5MS]

[SYS.02: GIS SPATIAL TRUTH [ALIGNED]]

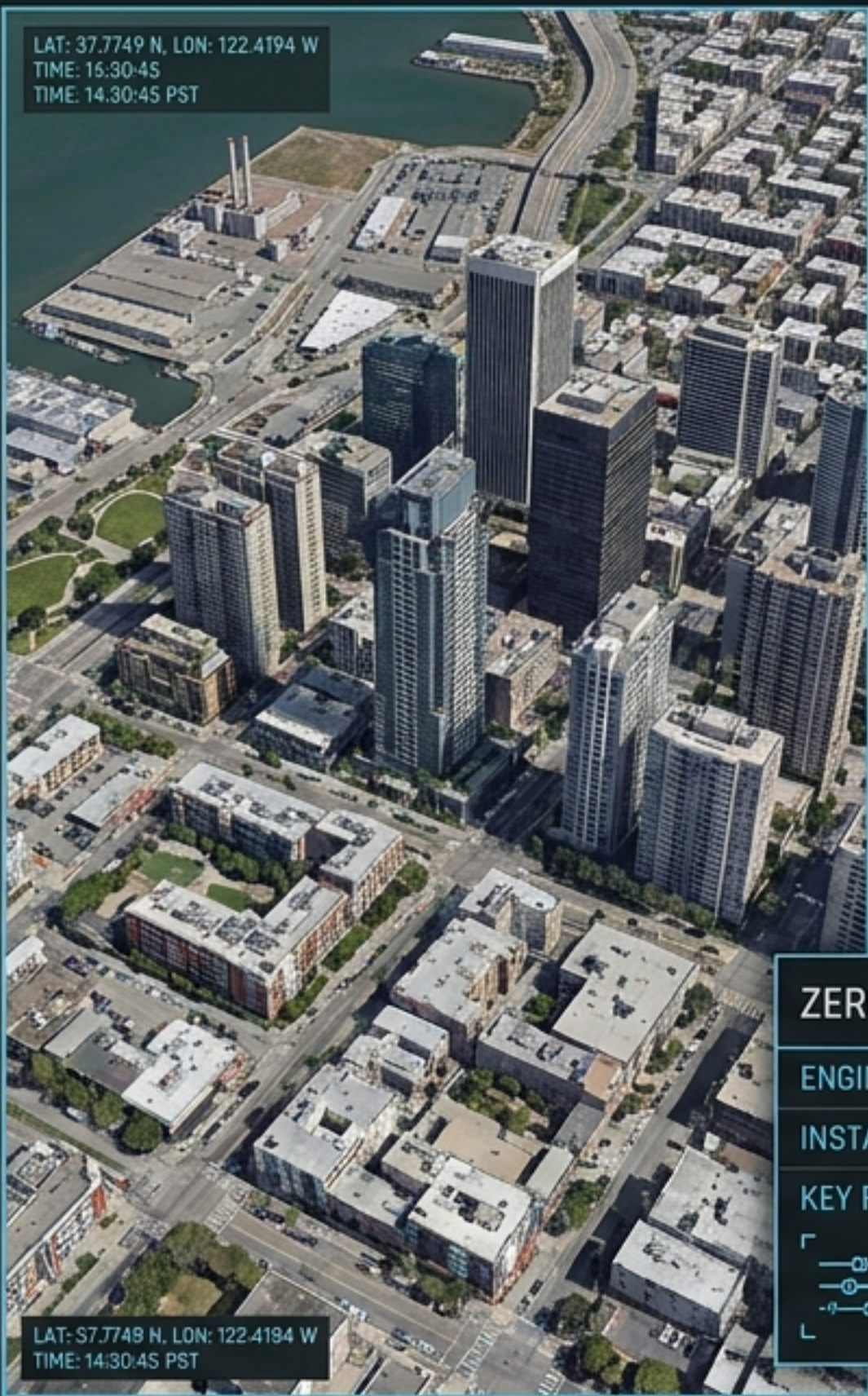
[STATUS: MAP_SYNC]
[LAYERS: 5 ACTIVE]
[ACCURACY: 99.9%]
[UPDATE_RT: LIVE]
[COORDS: N40°45'10" W73°59'38"]
[DATA_SOURCE: GEO_GRID]

[SYS.03: YOLO EDGE AI [ACTIVE]]

[STATUS: DETECTION_ON]
[OBJECTS_TRACKED: 256]
[ALERTS: 14 CRITICAL]
[MODEL: V8_NANO]
[CONFIDENCE: >95%]
[PROCESS_TIME: 22MS]
[THREAT_LEVEL: ELEVATED]

MODE: PHOTOREALISTIC SATELLITE

LAT: 37.7749 N, LON: 122.4194 W
TIME: 15:30:45
TIME: 14:30:45 PST



LAT: 37.7749 N, LON: 122.4194 W
TIME: 15:30:45
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MODE: THERMAL IRONBOW MAPPING

MAX TEMP: 65°C
AVG TEMP: 28°C
THERMAL CONTRAST: HIGH



MODE: WIREFRAME STRUCTURAL GHOSTS

STRUCTURAL LOAD: 45%
BEAM INTEGRITY: OPTIMAL
SUBSURFACE SCAN: ACTIVE



STRUCTURAL LOAD: 45%
BEAM INTEGRITY: OPTIMAL
SUBSURFACE SCAN: ACTIVE

ZERO-INSTALL 60 FPS FUSED REALITY RENDERING

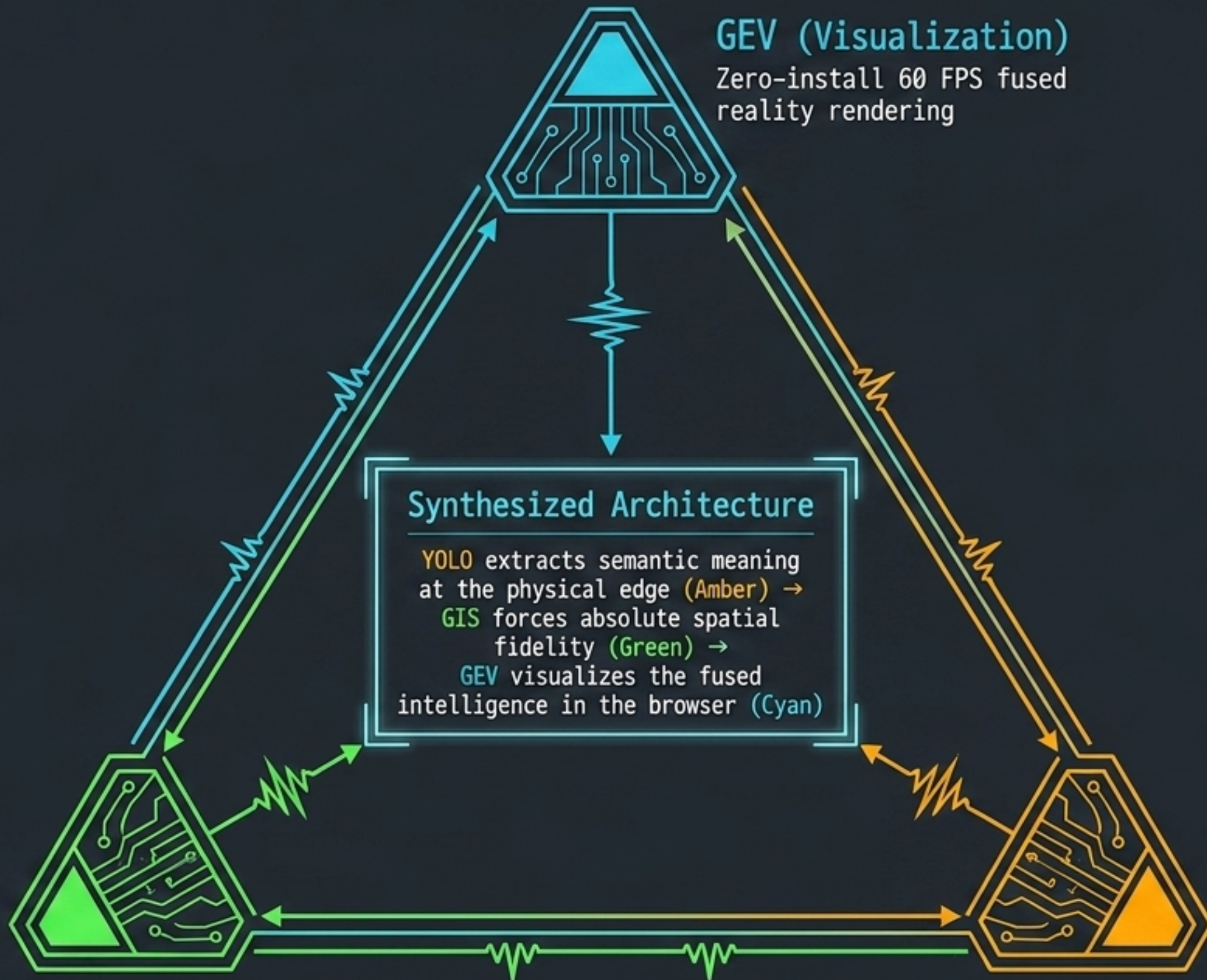
ENGINE: Browser-native WebGL2 / CesiumJS / Three.js

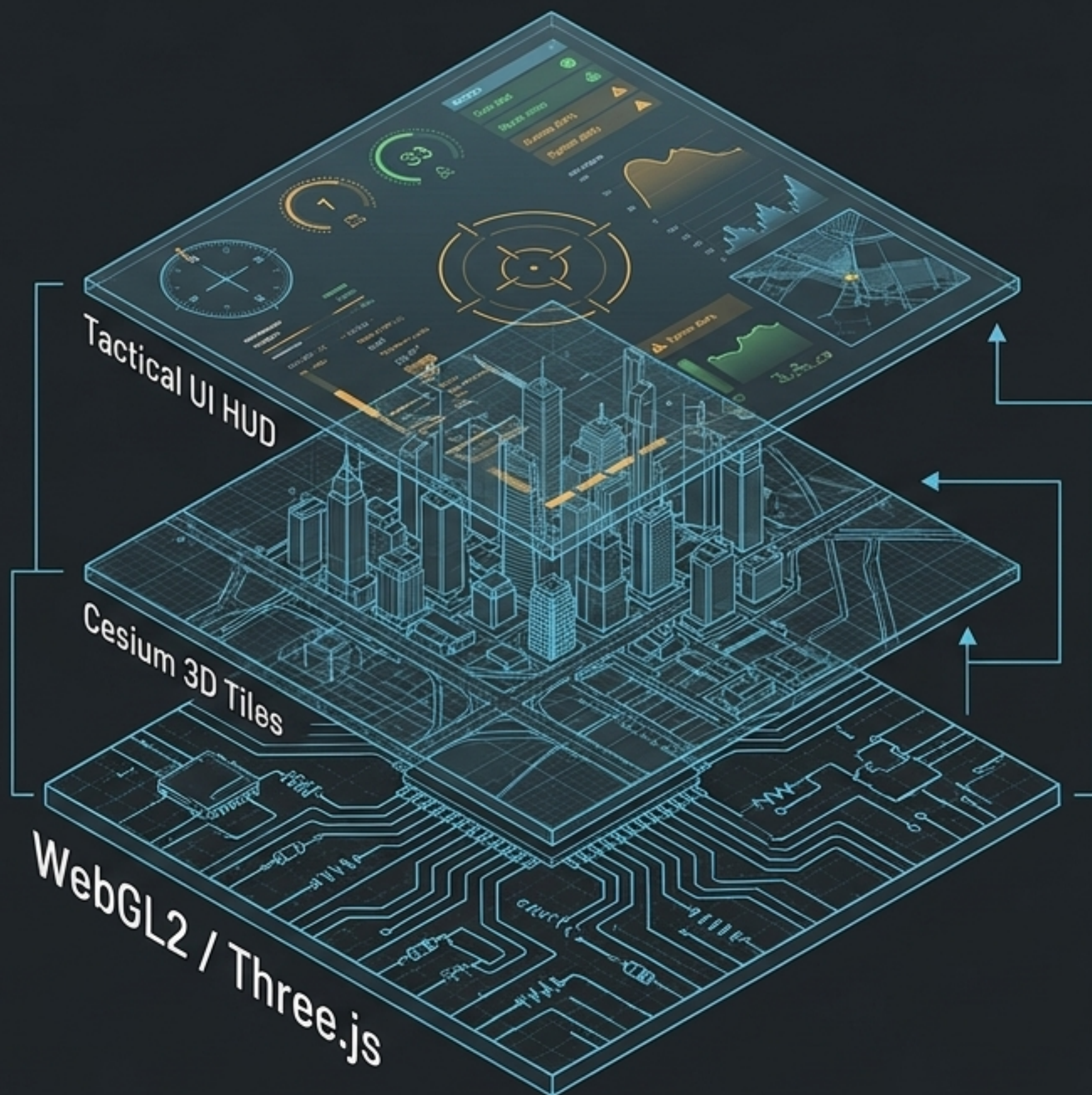
INSTALLATION REQUIREMENT: Zero (Client-side execution)

KEY FEATURES:

- Real-time live layers, cinematic framing, and world-stable icons utilizing per-frame screen-space course projection for accurate aircraft/vessel heading regardless of camera angle.

GIS (Spatial Truth)
Rigorous EPSG:4326
geodetic alignment





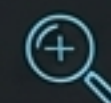
MEMORY PROFILING

Proximity-based 3D model swapping (assets render as 2D glyphs at macroscopic altitudes, swapping to high-fidelity glTF models on zoom).



GLYPH_RENDER: ACTIVE (ALTITUDE > 10KM)

GLTF_SWAP_THRESHOLD: 500M



VRAM_USAGE: 1.2GB [OPTIMIZED]

ASSET_THROUGHPUT: 120/s

SMOOTHING

Temporal telemetry interpolation (engine renders one interval behind real-time, using dead reckoning to smooth choppy 15-second data feeds).

DATA_FEED_INTERVAL: 15s

RENDER_DELAY: 1 INTERVAL

INTERPOLATION_FACTOR: 0.95

SMOOTHNESS_INDEX: 98% [STABLE]



SOVEREIGNTY

BYOK (Bring Your Own Key) model ensuring local-first execution. Proprietary data remains isolated in the client browser.



KEY_MANAGEMENT: LOCAL (CLIENT-SIDE)

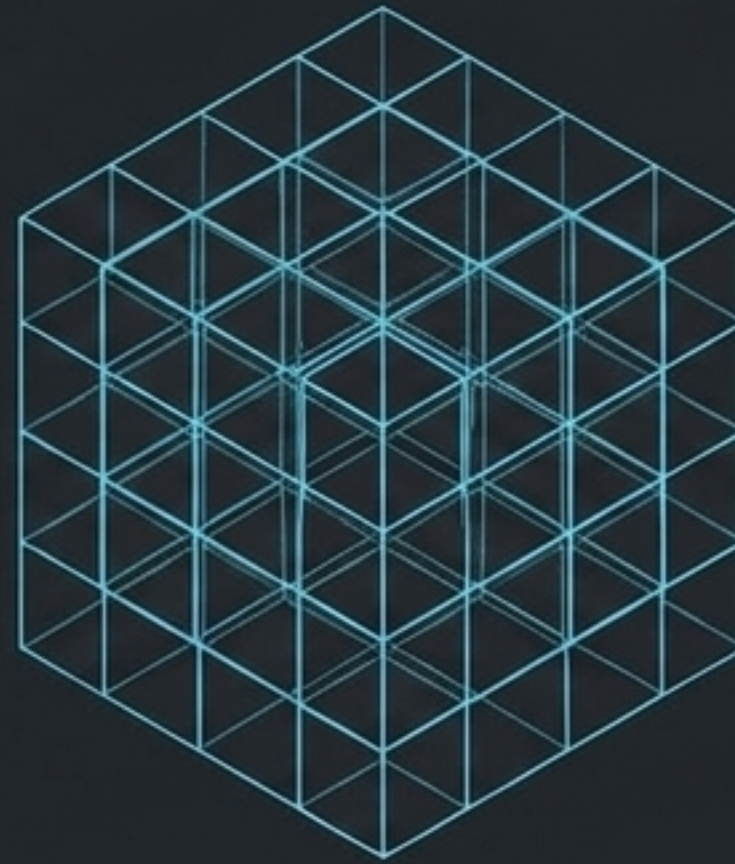
DATA_ISOLATION: ENFORCED



EXECUTION_ENVIRONMENT: BROWSER SANDBOX

ENCRYPTION_STATUS: AES-256 [SECURE]

WGS84 (EPSG:4326)

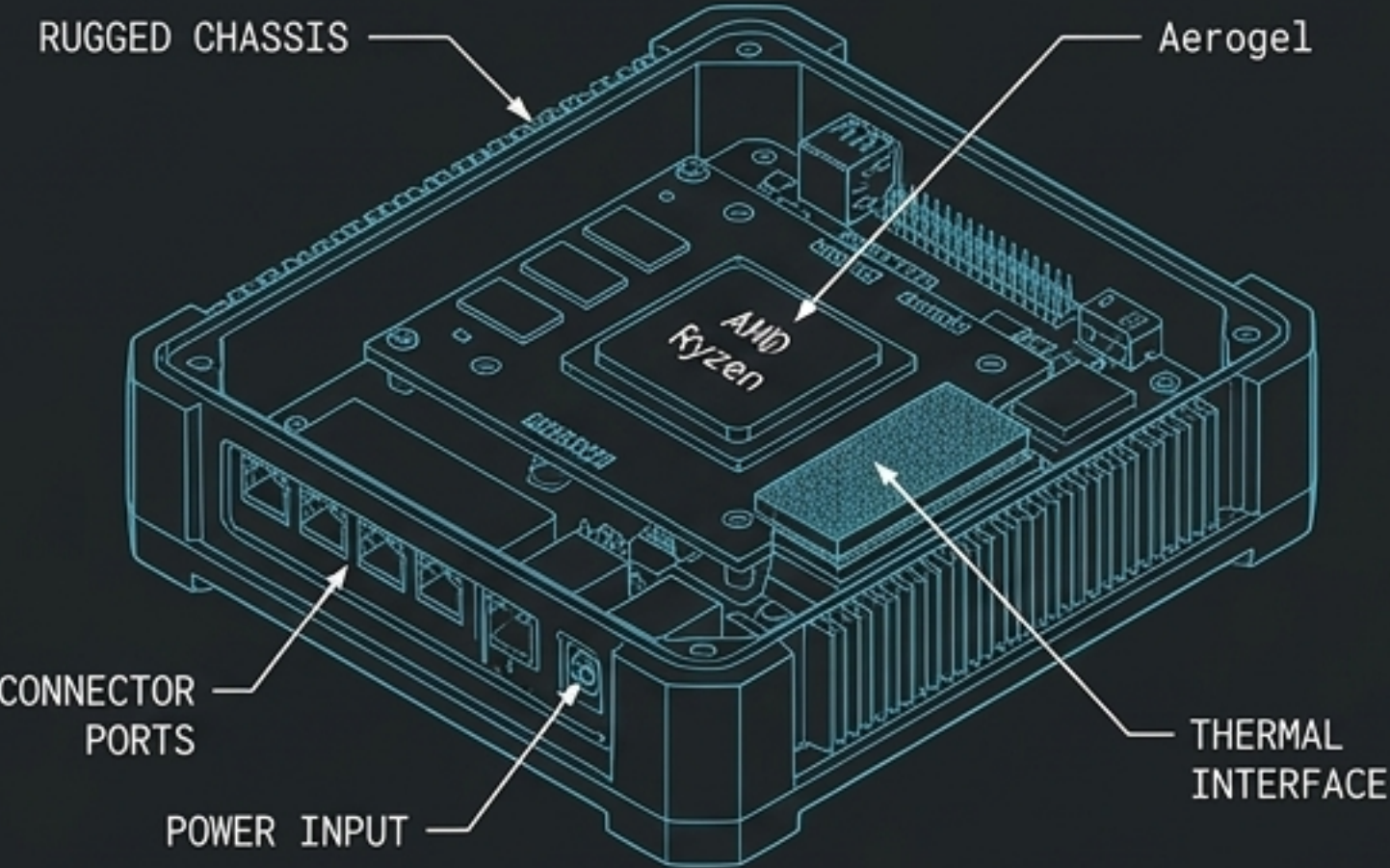
WGS84
(EPSG:4326)ECEF
(Earth-Centered, Earth-Fixed)ENU
(East-North-Up)

INSPECTOR

Anchoring Digital to Physical

Strict Coordinate Reference System (CRS) execution. All multi-modal inputs (satellite rasters, Sentinel-2 NDMI moisture maps, BIM infrastructure models, CCTV viewsheds) are mathematically forced into this unified alignment to ensure sub-meter spatial fidelity without drift.

HBK Mk-II HARDWARE NODE: SCHEMATIC OVERVIEW



CCTV FEED: LIVE YOLO INFERENCE



EDGE TELEMETRY READOUT

HARDWARE:
HBK Mk-II
Hydro-Bayesian Kit

THERMAL GOVERNANCE:
Passive_Survival Epistemic
Runtime rules (deterministic
capability downgrade under heat
stress rather than crashing)

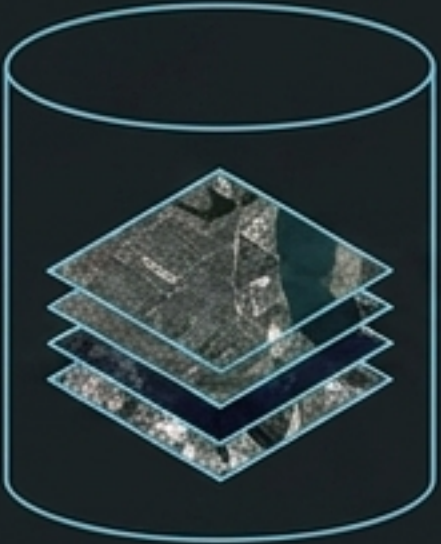
VISION COMPUTE:
Optimized, quantized YOLO
models (YOLOv5m/v8/11)
running locally at >270 FPS

OUTPUT:
Converts raw pixel video into
lightweight, semantic
variables sent via
deterministic telemetry packets

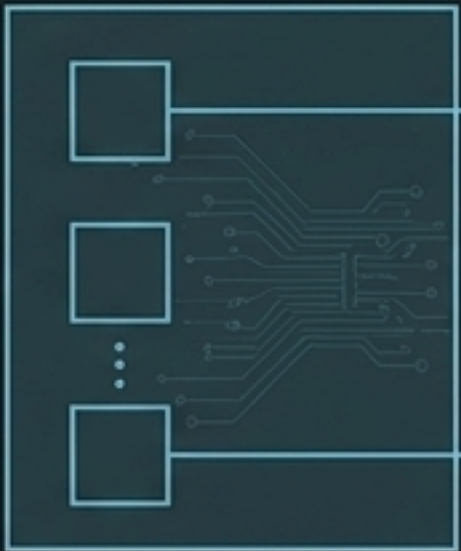
LAT: 97.7748 N, LON: 122.4184 W
LON: 220X
STATUS: 010X
TIME: 14:38:45 PST

SYSTEM STATUS: OPTIMA
SYSTEM STATUS: JK

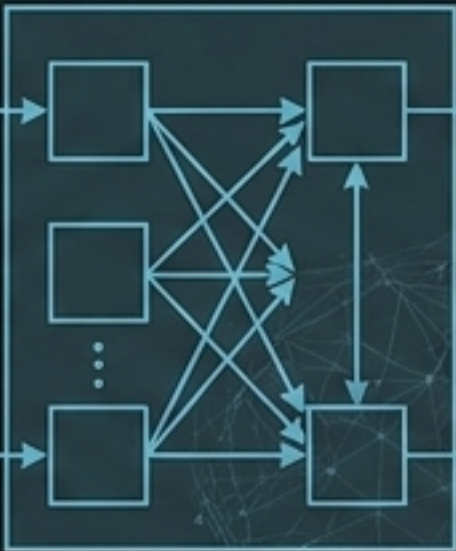
INPUT IMAGE DATASET



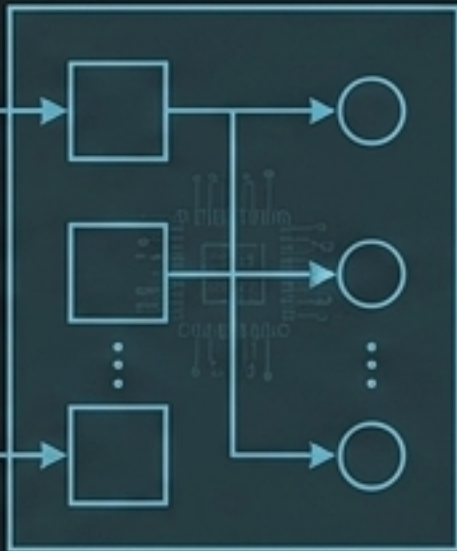
BACKBONE (Features Extraction)



NECK (Features Fusing)



HEAD (Features Detection)



OUTPUT IMAGE WITH PRECISE BOUNDING BOXES



ARCHITECTURAL EFFICIENCY

Replacement of CSP modules with novel C2f modules results in a lightweight framework. This structural shift drastically reduces parameter count and tensor size, enabling the edge node to process complex scenes instantaneously.

W0	235.67285	458.7089	W11	748753008
W1	233.20520	800.9090	W10	208106137
W2	20.23110	270.3070	W10	208.82307
W3	30.22100	320.2319	W11	88.00610
W4	0.10000	0.2000	W12	259150003
W5	139.90000	931.8500	W00	356317007
W6	233.30877	531.8902	W00	838100000
W7	0.00000	311.3333	W00	8888.8012
W8	0.50000	0.8000	W00	109150050
W9	0.20000	130.0000	W00	200.00002
W10	0.90111	318.2004	W00	0.0020001
W11	23.63757	0.2799	W00	370800222

LAT: 37.7749 N, LON: 38

W00: 00000000 000-0000 000 0000 00000000
W01: 00000000 000-0000 000 0000 00000000

W00: 00000000
W00: 00000000
W00: 00000000
W00: 00000000
W00: 00000000

SYSTEM TELEMETRY



ALGORITHM INSPECTOR

HMM NOISE FILTERING

	MECHANISM:	HIDDEN MARKOV MODEL (HMM) SEQUENTIAL FILTERING.
	LOGIC:	MODELS THE UNDERLYING STRUCTURAL STATE AS A HIDDEN VARIABLE (X_t) AND THE INSTANTANEOUS YOLO BOUNDING BOXES AS OBSERVATIONS (Y_t).
	RESULT:	SUPPRESSES TRANSIENT ENVIRONMENTAL ANOMALIES, DELIVERING ONLY STATISTICALLY VALIDATED OBJECT TRACKS AND PHYSICAL DEFECTS TO THE GEV ENGINE.

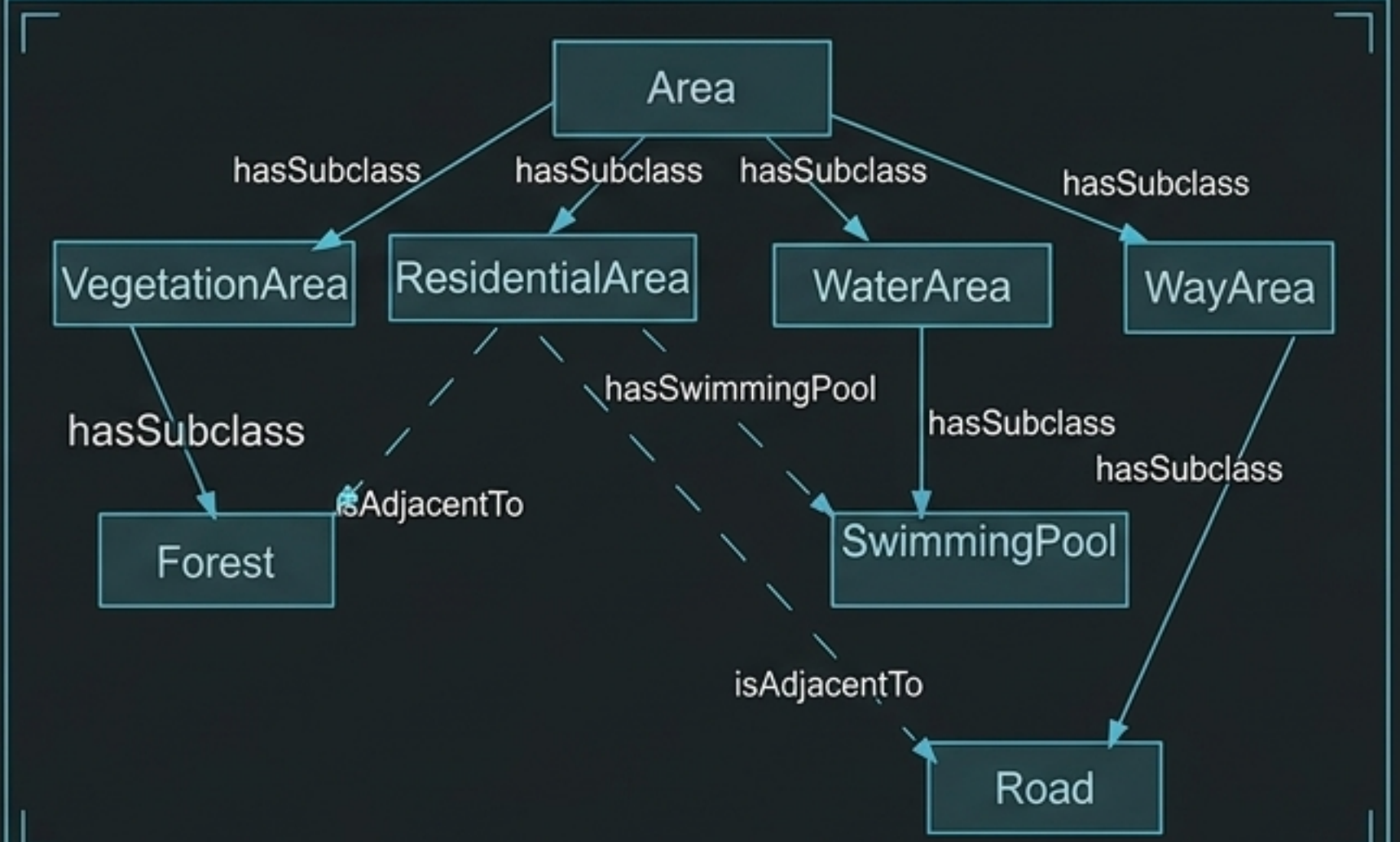
SEMANTIC REASONING & ONTOLOGICAL MAPPING ENGINE

LAT: 37.7749 N, LON: 122.4194 W | SYSTEM STATUS: OPTIMAL | TIME: 14:38:45 PST

RAW YOLO DETECTIONS (a)



ONTOLOGICAL NODE GRAPH (b)



SEMANTIC REASONING READOUT

THE PROBLEM:

A YOLO bounding box is blind to context. It simply sees **Water**.

THE SOLUTION:

The ontology maps formal subsumption relations.

DEDUCTION ENGINE:

Establishes spatial relationships (**isAdjacentTo**, **hasSwimmingPool**). By mapping **River** $\subseteq$ **WaterArea** (**Normal**) vs. **Pool** $\subseteq$ **WaterArea** near **Pipe** (**Anomaly**), the system logically deduces whether an edge detection is routine or a critical infrastructure alert.

VISUAL SENSOR DATA PROJECTION ENGINE

LAT: 37.7749 N, LON: 122.4194 W | SYSTEM STATUS: OPTIMAL | TIME: 14:38:45 PST

FLAT DASHBOARD CCTV

CAM ID: STREET_001

2024-05-15 14:38:45

REC



STREAM TYPE: RAW RTP H.264 | RESOLUTION: 1920x1080 | FRAMERATE: 30fps

VOLUMETRIC VIEWSHED CONE



GEOMETRY: WGS84 SPATIAL MESH | VIEWSHED SOURCE: CAM_STREET_001
PROJECTION: GLSL CUSTOM SHADER | COVERAGE ANALYSIS: ACTIVE

PROJECTION TELEMETRY PANEL

CUSTOM GLSL POST-PROCESSING

The GEV engine breaks out of flat dashboard containers by projecting live visual sensor data, thermal mappings, and Sentinel-2 NDMI/NMDI moisture rasters directly onto the WGS84 spatial geometry, revealing physical coverage and blind spots instantly.



ACTIVE SHADER
SHADER PIPELINE



TEXTURE MEMORY
USAGE: 2.63 TB



PROJECTION STATUS:
MAPPED

SYSTEM PERFORMANCE & RESOURCE UTILIZATION DIAGNOSTICS

LAT: 37.7749 N, LON: 122.4194 W | SYSTEM STATUS: OPTIMAL | TIME: 14:38:45 PST

VISUAL LAYER	EXECUTION PATH	JS HEAP	ACTIVATION	FRAME RATE
Idle Globe Baseline	Apple M5 Metal ANGLE	105.1 MiB	1,855.8 ms	60 FPS
CCTV City Activation	WebGL2 / Three.js	192.7 MiB	19,608.2 ms	60 FPS (48 live feeds)
YOLO Bounding Box (100%)	Dynamic Screen Overlay	124.8 MiB	817.6 ms	34.4 / 35.5 FPS (8,170 obs.)
Combined Static Stress	Multi-Layer Synthesis	872.2 MiB	30,000.0 ms	57.6 / 60 FPS (11,575 objects, 54,106 dynamic text draws)

SYSTEM CONCLUSION

The browser-native engine maintains near-60 FPS operability under massive spatial data synthesis.



OPERATIONAL USE CASE: VVU IVE VALIDATION

- **TARGET:** NON-REVENUE WATER (NRW) AND UNDERGROUND LEAKAGE.
- **DATA INGESTION:** THE SYSTEM INGESTS SPARSE SCADA OBSERVATIONS (FLOW, PRESSURE, PUMP STATUS).
- **HYDRAULIC DEVIATION TRIGGER:** BASELINE ESTABLISHES 96 L/s @ 48.6m. ANOMALY DETECTS 111 L/s @ 46.1m (FLOW INCREASED, PRESSURE DECREASED = POTENTIAL LOSS).
- **RESULT:** THE DIGITAL TWIN NARROWS THE SEARCH AREA FROM A WHOLE CITY TO A SPECIFIC DMA SEGMENT, TRIGGERING TARGETED FIELD VERIFICATION.

SYSTEM STATUS: ANOMALY DETECTED

LAT: 37.7749 N

LON: 122.4194 W

TIME: 14:40:05 PST

ALERT LEVEL: CRITICAL

Evidence Inflation (Bad) ⚠️

Scenario A (Without EIS):

5 pressure sensors on the same DMA drop simultaneously. System falsely claims 5 independent proofs = 100% confidence.

⚠️ FALSE CONFIDENCE

Scenario B (With EIS):

The system recognizes the 5 sensors are correlated. It caps them as 1 Primary + 1 Correlated event (Score: 0.5 - Insufficient).

SCORE: 0.5 - INSUFFICIENT

Independent Corroboration (Good) ✓

Scenario C (The Gold Standard):

1 Flow Anomaly (SCADA) + 1 Ground Moisture Report (Field) + 1 Acoustic Signal (YOLO Edge).

3 genuinely independent measurement principles.



EIS CALCULATION:

Primary (0.3) + Correlated (0.2) +
Independent (0.4) = 0.9 → VERIFIED_CANDIDATE

STATUS: VERIFIED_CANDIDATE



DATA SOVEREIGNTY & EXECUTION

BYOK Model: Epistemic Runtime rules dictate that all APIs are brokered server-side. Google Maps/Cesium keys remain isolated in the client's local browser. **Zero exposure** of proprietary municipal data.

SUBMISSION READINESS MATRIX

- ✓ INTERNAL MASTER REFERENCE: **READY** ✓
- ✓ RESEARCH COLLABORATION: **READY** ✓
- ⚠ MUNICIPAL TECHNICAL EVALUATION: **CONDITIONAL**
 - ✓ **RESOLVED BLOCKERS:** Assembly regeneration,
 - ✓ Design Freeze Level 1, KCL Parameter Library.
 - ⚠ **ACTIVE BLOCKERS:** Tolerance specification, Material selection, PCB definition.

SYSTEM STATUS: **OPERATIONAL**

LAT: 37.7748 N LON: 122.4194 W

TIME: 14:42:15 PST

ALERT LEVEL: **NOMINAL**

Executive Summary Readout

The platform successfully bridges raw physical environments and digital models.

METHOD: Rigorous `EPSG:4326` spatial registration + quantized `YOLO Edge AI` + Zero-install 60 FPS 3D visualization.

OUTCOME: Sparse municipal and infrastructure observations are correlated, independently assessed (`EIS v1.0`), provenance-tracked, and converted into auditable engineering evidence.

[AWAITING VALIDATION PROTOCOL INITIATION]